

Semiconductor Manufacturing Quality Analysis

Applying **Statistical Process Control (SPC)** and **Lean Six Sigma** tools to real production data from a semiconductor fabrication line, to find which process parameters drive product failures and whether the process is stable and capable.

 [Open In Colab](#)

Business question

A fab runs hundreds of in-line sensors, but engineers cannot watch all of them. **Which few parameters matter most for yield, and are they under control?**

Data

[UCI SECOM dataset](#): 1,567 production runs, 590 sensor measurements per run, each labelled Pass/Fail at final test (Jul–Oct 2008). The notebook downloads it automatically.

Approach

Step	Method	Quality tool
1. Data quality	Remove sensors that are mostly empty or constant	Measurement system check
2. Yield	Overall and weekly failure rate	Run chart
3. Root-cause screening	Rank sensors by gap between failed and passed runs	Pareto analysis
4. Stability	Control limits from moving range ($M\bar{R}/1.128$)	I-MR control charts
5. Capability	Compare first vs second half of production	Cpk
6. Action	Findings + recommendations	DMAIC mindset

Key results

- **Line yield: 93.4%:** 104 failed runs out of 1,567.
- **Data quality gap:** 144 of 590 sensors (24%) were unusable, mostly empty or constant, pointing to a data-collection problem in the fab.

- **Key sensors linked to failures:** S59, S103, S510.
- **Out-of-control runs fail 2–3× more often:**

Sensor	Out-of-control runs	Failure rate (out of control)	Failure rate (in control)
S59	129	13.2%	6.1%
S103	56	14.3%	6.4%
S510	57	19.3%	6.2%

- **Process capability (Cpk), first vs second half of production:**

Sensor	Cpk first half	Cpk second half	Trend
S59	0.65	1.69	Improved
S103	0.70	0.54	Worsened
S510	0.37	0.44	Not capable in either half

Recommendations: put S59, S103 and S510 on real-time SPC monitoring with alarms; investigate root causes of out-of-control runs (equipment, recipe,

material lot), starting with S510; repair data collection for sensors with mostly missing readings.

Assumptions & limitations

- SECOM does not publish specification limits, so Cpk uses **assumed limits** from passing runs in the first half of production.
- Sensor names are anonymised in the source data, so findings point to *which* signals to investigate, not the physical cause.
- This is association, not proven causation; confirming root cause would need process engineering input (DMAIC Analyze/Improve).

How to run

1. Click **Open in Colab** above (no installation needed).
2. Runtime → Run all.

Tools

Python · pandas · matplotlib · SPC · Pareto · Cpk ·
Lean Six Sigma

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*Data: McCann, M. & Johnston, A. (2008). SECOM. UCI
Machine Learning Repository.*